

PDN Abstraction for EMIR Feasibility — Research Directions

Contents

0. Phase status	2
1. Problem summary (from spec)	3
Spec pipeline (error loci)	3
2. What was built	3
2.1 Vendor extraction (black box)	3
2.2 Reduction framework	4
3. Root cause	4
3.1 Physical: grid-level R is not uniform	4
3.2 Current pattern modulates how visible the error is	4
3.3 Mathematical (still valid; now supporting, not the primary story)	5
4. What was tried	5
4.1 IR-drop fitting (implemented)	5
4.2 Localized Pixel-R (implemented)	5
4.3 Dual-layer / stub models (tried, not a successor)	6
4.4 Experiment matrix	6
4.5 Practical takeaway for PG prototyping	6
5. Assessment	6
6. Remaining research	7
6.1 Directions worth pursuing	7
6.2 What not to spend more time on	7
7. Suggested paper / thesis narrative	7
8. Related work	8
8.1 Closest to the remaining problem	8
8.2 Foundations	8
8.3 Chiplet / 2.5D context	9
8.4 Benchmarks	9
9. Open questions	9
10. Document history	9

Context: TSMC 3Dblox EMIR feasibility uses a single-layer Pixel-R model. The spec proposes automatic extraction of Pixel-R (R_x , R_y , R_z) from a real multi-layer PG by Kron-reducing the multi-layer admittance and matching eigenvalues of the reduced port matrix.

Status (internship): The root cause of the spec’s inaccuracy is identified. A spec-faithful reduction framework exists and can run IBM decks and the CoWoS-S SoC. Uniform Pixel-R plus eigenvalue least squares is not a viable feasibility model on realistic grids. Localized Pixel-R plus IR-drop fitting is the best *in-class* result. A better *model class* — a structured reduction of the Kron-reduced port matrix — is still open. Method progress beyond diagnosing the failure is limited.

Success criterion (“better”): Under **arbitrary** defined port voltages and currents, the abstracted model produces IR-drop results similar to the real (port-reduced) PDN.

Source spec: PDN_Abstraction_for_EMIR_Feasibility-TSMC_Spec.pdf

Central question (updated): Kron reduction at the chosen ports is exact. Error comes from (i) the uniform Pixel-R model class and (ii) fitting eigenvalues of G instead of the IR map. The remaining research problem is to reduce G'_M under a **structural constraint** (Pixel-R-like regular topology) with a controllable IR error bound.

0. Phase status

Phase	Original goal	Status
1. Case study + theory	Prove that eigen-fitting of uniform Pixel-R must err	Done. Physical root cause: grid-level equivalent R is not spatially uniform once metal pitch $\neq$ grid pitch. Theory: eigen LS $\neq$ IR; three global parameters cannot span G'_M .
2. Improve Pixel-R	Better analytical fit and/or localized Pixel-R, implemented	Partial. IR-vector fitting and per-grid / per-block (R_x, R_y, R_z) are implemented. Localized Pixel-R + IR fit is the best in-class result. Uniform Pixel-R still cannot represent cell-to-cell R variation.
3. Challenge Pixel-R	New model beyond single-layer Pixel-R	Started, no ceiling. Dual-layer (sparse + dense) meshes with voltage-source / current-sink stubs were tried. They do not overturn localized Pixel-R. The right successor problem is graph spectral sparsification of G'_M with a topology constraint .

Phase 1 → Phase 2 → Phase 3
 done partial open (retargeted)
 root cause IR + local structured sparsification of G'

1. Problem summary (from spec)

Item	Content
Today	Feasibility PDN = single-layer R-network with user-specified Pixel-R
Gap	Pixel-R is experience-based; need systematic extraction from real PG
Intended inputs	Chiplet PG (LEF/DEF/GDS or P&R), RC-tech, chip grid size, power net names
Variables	Pixel-R $x/y/z$ per power net
Assumption	Multi-layer and single-layer models share the same number of ports
Output	Optimal Pixel-R $x/y/z$

Spec pipeline (error loci)

1. Extract multi-layer R-network from PG.
2. Align ports: bumps/TSVs (voltage) + per-grid lumped current sinks.
3. Build MNA admittance; eliminate internals (Kron / Schur):

$$G'_M = G_P - G_C^\top G_I^{-1} G_C$$

4. Build single-layer Pixel-R admittance $G_S(R_x, R_y, R_z)$.
5. Fit by eigenvalue matching:

$$\min_i \sum_i (\lambda_i(G'_M) - \lambda_i(G_S))^2$$

using $G'_M = Q \Lambda^M_{diag} Q^T$ and the assumption $Q^T G_S Q = \Lambda^S_{diag}$.

Kron reduction (step 3) is **exact** at the chosen ports. Error is in **steps 4–5** (model class + fitting objective), with secondary error from **step 2** (port lumping / grid alignment). That split is confirmed by the internship case study.

2. What was built

2.1 Vendor extraction (black box)

Cadence Voltus and Synopsys RedHawk-SC Model Extraction reference flows collapse a multi-layer PDN into a **die model** stored as a SPICE netlist. How the tools form that netlist is not documented.

A PG-database input path was attempted and abandoned: SPICE could not be recovered from the database. The working framework therefore takes **SPICE netlists** only (IBM open-source decks; CoWoS-S SoC die model).

2.2 Reduction framework

Input SPICE $\rightarrow$ port lattice $\rightarrow$ Kron $G'_M \rightarrow$ abstract R-model $\rightarrow$ IR correlation. Three arms:

Arm	Model	Role
spec_flow	Uniform Pixel-R, one global (R_x, R_y, R_z)	Spec replica (eigen LS default; IR fit available)
local_flow	Same stars, per-block / per-cell (R_x, R_y, R_z)	Uniformity ablation
my_flow	Dual-ish meshes (tri-stagger / tri-square) with stubs at real pad/sink XY	Competing structured model

IBM cases run end-to-end. The CoWoS-S SoC runs; **IR-drop fitting is numerically usable, eigenvalue fitting is not** (numerical corruption of the eigen problem on that netlist).

Details of IR fitting as a current-weighted impedance-spectrum match are in [IR_FIT_WEIGHTED_SPECTRUM.md](#).

3. Root cause

3.1 Physical: grid-level R is not uniform

On the CoWoS-S testcase every metal layer is itself a **uniform** stripe grid, but **layer pitches differ** and the **Pixel-R grid pitch differs from those metal pitches**. After the die is cut into Pixel-R cells, each cell covers a different combination of stripes and vias, so the equivalent (R_x, R_y, R_z) seen by that cell is **not** the same as its neighbor's.

TSMC's spec experiment used a testcase where **every metal pitch and the grid pitch were identical**. Then every cell sees the same equivalent R , uniform Pixel-R is in the right model class, and eigen fitting looks good.

That mismatch is why feasibility and signoff IR **trends disagree** on CoWoS-S: the feasibility Pixel-R map is essentially **flat across the die**, while signoff retains spatial structure.

Uniform Pixel-R cannot represent cell-to-cell equivalent- R variation. This is the dominant, and unsurprising, failure mode.

3.2 Current pattern modulates how visible the error is

Stimulus	Effect
Uniform current (no hotspot)	Every cell is equally important. Spatial R error is fully visible; IR maps diverge from golden.
Non-uniform / floorplan current	A few eigenmodes of $Z = G^{-1}$ dominate. IR fitting can spend its few parameters on those modes; the IR <i>map</i> can look closer to golden even when the model class is still wrong.

This is the same statement as IR LS being a **C-weighted match of the impedance spectrum**, not a uniform match of $\text{spec}(G)$. See §4.1 and [IR_FIT_WEIGHTED_SPECTRUM.md](#).

3.3 Mathematical (still valid; now supporting, not the primary story)

Accuracy means $L(G_S) \approx L(G'_M)$ under mixed BC (pads voltage-fixed, sinks current-driven), **not** $\text{spec}(G_S) \approx \text{spec}(G'_M)$.

(i) **Eigenvectors are not shared.** Real symmetric A, B share an eigenbasis iff they commute. In general $G'_M G_S \neq G_S G'_M$, so forcing $Q^T G_S Q$ onto its diagonal wipes modal coupling.

(ii) **Eigenvalue LS $\neq$ IR.** $V = G^{-1}I$ (or the load Schur map). Matching $\sum_i (\lambda_i^M - \lambda_i^S)^2$ does not minimize $\|G_S^{-1}I - (G'_M)^{-1}I\|$. Matrices with the same spectrum and different eigenbases give different IR.

(iii) **Model-class capacity.** Uniform Pixel-R has $\dim \theta \leq 3$. After Kron, G'_M is a dense port Laplacian, $\Theta(n^2)$ degrees of freedom. Nearest-neighbor stars cannot close

$$\lim_{\theta \rightarrow 0} \sum_j \left\| V_S(\theta; I_j) - V_M(I_j) \right\|_2^2$$

for large n . Localization raises the dimension (one triple per cell) and is why Phase 2 helps *inside* the star family — it still does not recover long-range Kron fill-in.

(iv) **Mixed BC.** The operator that should be fitted is Z_{load} (sink Schur given fixed pads), not free $\text{spec}(G)$.

Secondary: per-grid current lumping; grid pitch $\gg$ bump pitch (the spec already flags this).

4. What was tried

4.1 IR-drop fitting (implemented)

Replace eigenvalue LS with a direct IR-vector loss on mixed-BC sink voltages:

$$\min_{\theta} \sum_j \left\| V_S(\theta; I_j) - V_M(I_j) \right\|_2^2 = \min_{\theta} \sum_j \left\| (Z_S(\theta) - Z_M) I_j \right\|_2^2.$$

With training Gram $C = \sum_j I_j I_j^T$ this is Hilbert–Schmidt approximation of Z on $\text{range}(C)$. Limited parameters are spent on Laplacian modes **the current pattern actually excites**, and (because $V = Z I$) on small λ (global IR) rather than uniformly on $\text{spec}(G)$.

On CoWoS-S this is also the only numerically stable objective; eigen LS corrupts.

4.2 Localized Pixel-R (implemented)

Keep the star topology; let (R_x, R_y, R_z) vary by $K \times K$ pad block (local flow: $K=0$ global, $K=1$ per cell). Shared edges use series half-arms. This is the in-class fix for §3.1.

Diagnostic snapshot (IBM, seed 0, auto pitch, vs uniform Pixel-R IR fit):

Case	K	regions	$e_{\text{IR}} \text{ localized}$	$e_{\text{IR}} \text{ uniform}$
ibmpg2	0	1	0.043	0.043
ibmpg2	2	25	0.0063	0.043
ibmpg2	1	81	~0	0.043

Case	K	regions	e_{IR} localized	e_{IR} uniform
ibmpg4	0	1	0.104	0.104
ibmpg4	1	225	0.028	0.104

Held-out random / localized / striped stimuli improve with $K=1$ as well; the gain is not only overfitting the training current. **Localized Pixel-R + IR fitting is the best result in this project.**

4.3 Dual-layer / stub models (tried, not a successor)

my_flow attaches pads and sinks at **physical** C4 / current-sink XY through short stubs, and uses two mesh sheets (one coarser, one finer). This restores some layer-dependent R that single-layer Pixel-R erases. It is a legitimate feasibility-style topology. It has **not** been shown to beat localized Pixel-R on IR under arbitrary V/I , and it does not solve the structural-sparsification problem in §6.

4.4 Experiment matrix

Experiment	Vehicle	Role
Non-uniform current	CoWoS-S	Matches floorplan; IR map closer to golden
Uniform current	IBM	Amplifies spatial R error (no hotspot)
Random perturbation of grid-cell R	IBM	Stresses uniformity assumption
Localized Pixel-R (independent per-grid R_x, R_y, R_z)	IBM	Best in-class
Finer pitch (cells need not contain a voltage source)	IBM	Grid vs bump occupancy
Dual-layer sparse+dense + VS/sink stubs	IBM	Phase-3 prototype

4.5 Practical takeaway for PG prototyping

If die size and Pixel-R grid size are chosen **while the PG is being built**, so that each cell covers a **repeated / similar** stripe pattern (ideally metal pitch commensurate with grid pitch), uniform Pixel-R is much closer to the right model class. That is an implementation guideline, not a reduction theorem.

5. Assessment

Claim	Verdict
Uniform Pixel-R cannot capture grid-level equivalent- R differences	Confirmed; obvious once metal pitch $\neq$ grid pitch
The spec testcase hid this by using matched pitches	Confirmed on CoWoS-S vs spec experiment
Eigen LS is the wrong objective for IR	Confirmed; also numerically unusable on CoWoS-S
IR fitting + localized Pixel-R is the best in-class method here	Confirmed on IBM; IR arm is the usable CoWoS-S path
A new model that overturns Pixel-R with a better accuracy/structure tradeoff	Not achieved

The internship **clarified the failure**. It did **not** produce a new abstraction with a theory of error that feasibility tools can rely on. Phase 2 is an engineering mitigation (more Pixel-R parameters + the right loss).

Phase 3 remains the research problem.

6. Remaining research

After Kron, the exact port model is the (generally dense) matrix G'_M . Building a Pixel-R-like mesh is asking for a **sparse, highly structured** G_S such that the mixed-BC IR map of G_S tracks that of G'_M .

Unconstrained, that is **graph spectral sparsification** (and related effective-resistance / Schur sparsifiers). Existing work (e.g. Spielman–Srivastava; Liu–Yu pGRASS and ICCAD’23 multilevel aggregation) sparsifies for solve accuracy or IR on the reduced *graph*, but **does not constrain the reduced topology** to a regular Pixel-R (or dual-layer mesh) stencil.

A feasibility Pixel-R is exactly G_S in a **structured subset** of Laplacian matrices: nearest-neighbor stars on a 2-D lattice, optionally with a second sheet and vias; parameters are edge weights in that stencil only. Little published work treats spectral sparsification **under that structural constraint**.

6.1 Directions worth pursuing

1. **Structured sparsification.** Approximate G'_M (or Z_{load}) by a Pixel-R / dual-mesh Laplacian. Questions: which matrix distance is IR-relevant; how to project a general Schur complement onto a stencil; whether localization is the *optimal* stencil-weight assignment or only a greedy one.
2. **Error bounds.** Analytic bounds on $\|Z_S I - Z_M\|$ (or e_{IR}) in terms of pitch mismatch, stencil bandwidth, and the current Gram C . The internship’s qualitative claims (uniform I is worst; commensurate pitches help) should become theorems or sharp examples.
3. **Prototyping-aware grids.** Treat grid pitch as a design variable so cells see a periodic PG unit. This may be the highest-leverage *engineering* fix; it does not replace (1)–(2).
4. **Krylov / structured MOR** (optional). Compact multiport models with moment guarantees, if the feasibility API may leave a mesh. Not the lead bet: 3Dblox still wants a Pixel-R-shaped R-network.

6.2 What not to spend more time on

- Re-deriving that eigen LS $\neq$ IR.
 - Further uniform- (R_x, R_y, R_z) tuning on mismatched-pitch CoWoS-S.
 - Treating vendor die-model generation as a research target (black box; SPICE-in is enough).
-

7. Suggested paper / thesis narrative

1. **Problem.** Multi-layer PDN $\rightarrow$ single-layer Pixel-R for 3Dblox EMIR feasibility; need IR fidelity under arbitrary port V/I .
 2. **Baseline.** TSMC eigen-fitting of uniform Pixel-R to Kron-reduced G'_M . Looks good when metal pitch = grid pitch; fails on CoWoS-S (flat feasibility IR vs structured signoff).
 3. **Diagnosis.** Cell-level equivalent R varies after gridding; three global parameters cannot represent it. Eigen LS matches the wrong operator; on CoWoS-S it is also numerically unstable.
 4. **In-class mitigation.** IR-vector fitting (current-weighted Z) + localized Pixel-R. Best empirical result; still a star mesh.
-

5. Open problem. Spectral sparsification of G'_M with a **Pixel-R topology constraint**, plus IR error bounds. Optional: dual-layer stencil, Krylov MOR if the API may change.

Metric: e_{IR} / e_{worst} on held-out stimuli — not eigenvalue residual.

8. Related work

Prioritize sparsification **with structure**, then Kron / R_{eff} , then MOR.

8.1 Closest to the remaining problem

Year	Paper	Why
2023	Z. Liu, W. Yu, <i>pGRASS-Solver</i> , IEEE TCAD	Spectral sparsification + effective resistance for large grids. Gap: no Pixel-R stencil constraint. IEEE · PDF
2023	Z. Liu, W. Yu, <i>Accuracy-Preserving Reduction of Sparsified Reduced Power Grids...</i> , ICCAD	Schur / sparsified reduced grids + IR. Hierarchical, still not a regular feasibility mesh. IEEE · PDF
2011	D. Spielman, N. Srivastava, <i>Graph Sparsification by Effective Resistances</i> , SIAM J. Comput.	Classical spectral sparsifiers; unconstrained support. SIAM
2024	A. Carlucci et al., <i>Structured Model Order Reduction of System-Level Power Delivery Networks</i> , IEEE Access	Krylov / structured MOR — optional Phase 3 if the API may leave a mesh. IEEE

8.2 Foundations

Year	Paper	Why
2013	F. Dörfler, F. Bullo, <i>Kron Reduction of Graphs...</i> , IEEE TCAS-I	Exact port Schur; R_{eff} invariance. IEEE
2023	T. Sugiyama, K. Sato, <i>Kron Reduction and Effective Resistance of Directed Graphs</i> , SIAM J. Matrix Anal. Appl.	Modern Kron / R_{eff} . SIAM · arXiv
2016	Abed, Najm et al., <i>A Fast Layer Elimination Approach for Power Grid Reduction</i> , ICCAD	Layer thinning → dual-layer stencils. PDF
2019	S. Köse, E. Friedman, <i>Effective Resistance of Two-Dimensional Truncated Infinite Mesh Structures</i> , IEEE TCAS-I	Mesh R_{eff} . IEEE

8.3 Chiplet / 2.5D context

Year	Paper	Why
2025	<i>PDN analysis of 3D chiplet integration...</i> , IEICE ELEX	Chiplet PDN + IR. DOI
2024	X. Ma et al., <i>Electrical-Thermal Co-Simulation of Chiplet Heterogeneous Integration</i> , IEEE TVLSI	Chiplet PDN equivalents. IEEE
2023	A. Carlucci et al., <i>Compressed Multivariate Macromodeling...</i> , IEEE TCPMT	Response-preserving macromodels vs eigen-only fit. IEEE

8.4 Benchmarks

IBM PG (this repo) and SRAM-PG (Shen, Liu, Yu, 2024, [arXiv:2404.05260](https://arxiv.org/abs/2404.05260)). CoWoS-S die-model netlist is local / vendor, not public.

9. Open questions

1. What IR error does 3Dblox feasibility actually tolerate (mV vs %)?
2. May the feasibility API emit **localized** (per-cell) Pixel-R, or must output stay one global (R_x, R_y, R_z)?
3. Can a Pixel-R / dual-mesh stencil be posed as a convex or provably near-optimal sparsifier of G'_M or Z_{load} ?
4. Error bound in terms of pitch mismatch and current Gram C ?
5. Ports = bumps + grid sinks only, or instance-level currents later?
6. Static IR only, or RC / dynamic EMIR (would favor Krylov MOR)?

10. Document history

Date	Note
2026-07-21	Initial consolidation from TSMC PDN Abstraction spec analysis.
2026-07-21	Merged official research plan: 3 phases; commutativity / over-simplification / $G^{\sim}I$ proofs; physics-response fitting; localized Pixel-R; dual-layer model; Krylov MOR.
2026-09-01	Rewritten from internship results: CoWoS-S pitch/grid root cause; SPICE-only framework; IR fit + localized Pixel-R as best in-class; Phase 3 retargeted to structured spectral sparsification of G'_M .